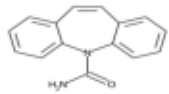


Paired Dataset

Drug structure

SMILES -> ECFP4



matched pair

Mechanism text

sodium channel
beta-2 agonist
DNA polymerase

Frozen Encoders

ECFP4 / RDKit

2048-bit fingerprint

S-Biomed-RoBERTa

CLS from each layer

Trainable Bridges

molecule bits
feature map

trainable
projection heads

W_m

2048 -> 256

text states
feature map

W_t

768 -> 256

Shared Retrieval Space

CLOOB / InfoLOOB objective

align matched pairs; repel impostors

supervises alignment

256-d joint space

cosine / dot-product retrieval

● molecule ● text ● anchor

$$s_{ij} = \hat{b}_i^{(m)} \cdot \hat{b}_j^{(t)}$$

Audit Outputs

explain

ECFP4 saliency



retrieve

Ranked retrieval



probe

RoBERTa logit lens



Legend

● molecule branch

● text branch

■ frozen encoder

■ trainable bridge

■ shared space

■ loss objective